BIO-301 FINAL TERM- 100%

GUESS PAPER 2023

Essentials of Genetics

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SUBJECTIVES

VNTR and its applications:

VNTR are short nucleotide sequences organized as a tandem repeats on genomes. Found on many chromosomes, Show variations in length between individuals.

Application: 1.DNA fingerprinting 2.genetic diversity 3. forensic mapping of genome 4. breeding pattern of wild and domesticated animals 5. paternity testing 6. Evolutionary research .

Procedure for western blotting:

The Western blotting procedure involves several steps. First, proteins are separated by size using gel electrophoresis. Then, the proteins are transferred onto a membrane. The membrane is blocked to prevent non-specific binding, and then incubated with a primary antibody that binds to the target protein. After washing, a secondary antibody is added that recognizes the primary antibody. Finally, the protein bands are visualized using a detection method such as chemiluminescence.

Effect of environment on allelic expression:

Allele expression is influenced by environmental conditions, as seen in arctic foxes, Himalayan rabbits, and Siamese cats. For instance, the ch allele in Siamese cats is impacted by temperatures above 33°C, causing inactivation of the tyrosinase enzyme, leading to reduced melanin pigment and resulting in their characteristic coat color patterns.

Explain trisomy and it's Types

Trisomy is a type of polysomy in which there are three copies of a particular chromosome,instead of the normal two. Trisomy is a type of aneuploidy.

Types:

1. **Trisomy 21 (Down syndrome):** Caused by an extra copy of chromosome 21, leading to intellectual disabilities and distinct facial features.
2. **Trisomy 18 (Edwards syndrome):** Resulting from an extra chromosome 18, it causes severe developmental issues and organ abnormalities.
3. **Trisomy 13 (Patau syndrome):** Arising from an extra chromosome 13, it causes serious birth defects and often shortened lifespans.

Difference between Ring and Philadelphia chromosome:

1= A ring chromosome is a chromosome whose arms have fused together to form a ring. In humans it is denoted by small r and in drosophila it is denoted by capital R., **2= Philadelphia chromosome:**- is a specific chromosomal abnormality that is related with chronic myelogenous leukemia(CML). Reciprocal translocation between chromosome 9 and chromosome 22 which is specifically designated.

Types of chromosome on the base of centromere location:

1. **Metacentric:** Centromere centrally located, resulting in equal arms.
2. **Submetacentric:** Centromere slightly off-center, leading to unequal arms.
3. **Acrocentric:** Centromere near one end, creating a longer and a shorter arm.
4. **Telocentric:** Centromere at the end, resulting in a single long arm.

Effects of point mutation.

Point mutations alter single DNA bases, affecting protein structure/function. Substitutions (missense, nonsense) change amino acids or stop codons, impacting gene expression and possibly causing disease or genetic diversity.

How giemsa stain work/ Define giemsa stain.

Geimsa stain is a type of stain used in microscopy to visualize and differentiate different types of cells, especially blood cells. It works by Giemsa stain binds to DNA and RNA, creating distinct bands on chromosomes, allowing for better visualization of cellular structures and identification of specific cell types.

RAPD,ALFP difference:

RAPD: A PCR-based technique used to amplify random segments of genomic DNA using single primers. The amplified products reveal polymorphisms between individuals. Generally less expensive and simpler, as it requires fewer primer pairs.

AFLP: A PCR-based method that selectively amplifies subsets of restriction enzyme-digested genomic DNA, creating a fingerprint of fragments that differ among individuals. Usually more expensive and complex, involving multiple primer pairs and enzyme digestion steps.

Write about the non.disjunction/ Caused by ?

Non-disjunction is a genetic event that can occur during cell division, specifically during the separation of chromosomes. It can be caused by various factors, including genetic abnormalities, environmental factors, and maternal age. Non-disjunction can lead to an abnormal number of chromosomes in the resulting cells, which can cause genetic disorders such as Down syndrome.

Erythrocytes lysis in RNA purification:

In RNA purification, erythrocyte lysis is a step where red blood cells are broken open to release their contents. This is usually done using a lysis buffer that disrupts the cell membranes and releases the RNA. The lysate is then processed further to purify the RNA of interest.

Linked and causative single polymorphism.

A linked single nucleotide polymorphism (SNP) is located near another genetic variant on the same chromosome, often inherited together. A causative SNP directly influences a trait or disease, demonstrating a functional relationship through its impact on gene expression or protein structure.

10 organism names with chromosomes:

1= Human: 46 chromosomes, 2= Mouse: 40 chromosomes, 3= Fruit Fly: 8 chromosomes, 4= Dog: 78 chromosomes, 5= Cat: 38 chromosomes, 6= Rice: 24 chromosomes, 7= Corn: 20 chromosomes, 8= chimpanzee :48 chromosomes, 9= Elephant: 56 chromosomes, 10= Tomato: 24 chromosomes.

Downstream genetic mapping:

Downstream genetic mapping refers to the process of analyzing genetic data to identify and map specific genes or genetic markers that are associated with a particular trait or disease. This involves conducting statistical analyses, such as genome-wide association studies (GWAS), to identify regions of the genome that are linked to the trait of interest. This information can then be used to further investigate the underlying genetic mechanisms and potential therapeutic targets.

Patterns of genetic disorders:

Patterns of genetic disorders refer to recognizable ways in which genetic conditions are inherited within families. These include:

1. **AUTOSOMAL DOMINANT:** Affected individuals are heterozygote. One allele is mutated, while other one is normal.
2. **AUTOSOMAL RECESSIVE:** Homozygous, when both alleles of the gene are mutated.
3. **X-LINKED RECESSIVE:** Mutated alleles are present on either of the sex chromosomes.
4. **MULTIFACTORIAL:** Genes and environment both contribute.

Hot Start PCR:

Hot Start PCR is a modified PCR technique that prevents non-specific amplification and enhances specificity. It involves initially deactivating the DNA polymerase enzyme at lower temperatures. This prevents early DNA replication until the reaction is heated to the optimal temperature, reducing false products. Hot Start PCR is preferred over conventional PCR to improve target amplification accuracy. It's useful for complex DNA samples with multiple similar sequences, minimizing unwanted amplification. Hot Start PCR is crucial in applications like genotyping, DNA sequencing, and analyzing low-quality samples where specificity is crucial for reliable results.

Diagnostic test for genetic disorder:

A diagnostic test for a genetic disorder usually involves analyzing a person's DNA to detect specific mutations or variations associated with the disorder. These tests can include techniques like PCR, DNA sequencing, and microarray analysis. Keep in mind that specific tests depend on the suspected disorder. It's important to consult a medical professional for accurate guidance and testing recommendations.

Tests:

PCR-Based Tests, DNA Sequencing Fluorescence , In Situ Hybridization (FISH), Microarray Analysis, Southern Blotting, Northern Blotting, Western Blotting, Karyotyping, Cytogenetic Analysis, Genomic DNA Methylation Analysis, Whole Exome Sequencing (WES).

Applications of PCR or Uses of PCR:

PCR has applications in DNA amplification, 1.Genetic testing 2 forensics, medical diagnostics, 3.Environmental studies 4.Evolutionary studies 5.food safety 6 .Archaeology, and anthropology.

Define interphase and cell cycle stages 3

Interphase: is the stage in the cell cycle where the cell prepares for division. It is divided into three phases: G1 (Gap 1), S (Synthesis), and G2 (Gap 2). During interphase, the cell grows, carries out its normal functions, and replicates its DNA in the S phase. The cell cycle stages include interphase and mitosis (or meiosis in reproductive cells).

Principle of pCR:

The thermocycler works on the principle of Peltier effect, which raises and lowers the temperature of the block in a pre-programmed manner.

The principle of PCR (Polymerase Chain Reaction) involves the enzymatic amplification of a specific DNA sequence. It consists of repeated cycles of heating (denaturation), cooling (annealing), and DNA synthesis (extension). This process uses a DNA template, DNA primers,

and a heat-stable DNA polymerase to exponentially produce copies of the target DNA region, facilitating various genetic analyses.

Interphase and its events LONG:

Interphase is a phase of the cell cycle where the cell prepares for division. It includes three events: G1 phase (cell growth and metabolic activity), S phase (DNA replication), and G2 phase (further growth and preparation for division). During interphase, the cell synthesizes proteins, duplicates its DNA, and ensures all components are ready for subsequent cell division, either mitosis or meiosis.

CAUSES OF EPIGENETICS:

1= DNA Methylation, 2= Histone Modification, 3= miRNAs.

Use of mgcl2 and EDTA:

Magnesium chloride (MgCl2) is used in molecular biology reactions, like PCR, as a cofactor for DNA polymerase activity, enhancing enzyme efficiency. Ethylene diamine tetra acetic acid (EDTA) is employed to chelate divalent metal ions like Mg2+, inhibiting DNases and RNases, preserving DNA and RNA integrity.

Deletion mutation:

A deletion mutation is a genetic alteration where a segment of DNA is lost, leading to the removal of one or more nucleotides. This can cause frameshifts, affecting protein synthesis and potentially causing genetic disorders.

Causes: Replication errors, DNA strand slippage, Radiation or mutagen exposure, Chemical damage, Errors during recombination.

Missense mutations and its effect on protein formation:

Missense mutations are genetic alterations where a single nucleotide change in DNA leads to a different amino acid in the protein sequence. This can alter protein structure, function, and interactions, affecting its stability, catalytic activity, or binding capacity, potentially leading to malfunction, disease, or altered traits.

Environmental and genetic causes of variation.

Variations in traits can be caused by both environmental and genetic factors. Environmental factors include things like diet, exposure to toxins, and physical activity, which can influence gene expression and affect traits. Genetic factors, on the other hand, involve variations in DNA sequences that are inherited from parents and can directly impact traits. It's important to note that in some cases variations are a result of a combination of both environmental and genetic influences.

Difference between hot star and suicide PCR:

Hot Start PCR is a modified PCR technique that prevents non-specific amplification and enhances specificity. It uses modified DNA polymerases that activate after an initial denaturation step, reducing background noise and improving target amplification.

Suicide PCR, also known as "Touchdown PCR," is used to optimize annealing temperatures for challenging templates by gradually reducing the temperature in successive cycles. This improves specificity, reduces nonspecific amplification, and enhances target amplification efficiency.

Northern western southern blot difference:

Southern blotting:

Southern blot is a method used in molecular biology for detection of a specific DNA sequence in DNA samples.

Northern blotting:

The northern blot is a technique used in molecular biology research to study gene expression by detection of RNA (or isolated mRNA) in a sample.

Procedure of southern blotting:

1. DNA Digestion: The DNA sample is cut into fragments using restriction enzymes. **2. Gel Electrophoresis:** Fragments are separated by size using gel electrophoresis. **3. Denaturation:** DNA fragments are treated to separate the double-stranded DNA into single strands. **4. Transfer:** Fragments are transferred from the gel to a membrane, like nitrocellulose or nylon. **5. Hybridization:** A labeled DNA probe binds to the target DNA sequences on the membrane. **6. Washing:** Non-specifically bound probes are washed away, leaving only the hybridized ones. **7. Detection:** The presence of the labeled probe is visualized using autoradiography or other detection methods. This technique is used to detect specific DNA sequences and analyze DNA fragment sizes.

Define leading strand, Lagging strand, okazaki fragments. Gyrase and sister chromatids

In DNA replication, the **leading strand** is synthesized continuously in the 5' to 3' direction, following the replication fork, while the **lagging strand** is synthesized discontinuously in short fragments called Okazaki fragments. **Okazaki fragments** are later joined by DNA ligase to form a continuous strand. **Gyrase** is an enzyme that helps relieve the tension caused by DNA unwinding during replication. **Sister chromatids** are two identical copies of a chromosome joined together at the centromere, formed during DNA replication, and are separated during cell division.

HGCN rule for substitution, range of residue duplication deletion insertion inversion

The HGCN rule, or the Human Genome Variation Society nomenclature, provides guidelines for describing genetic variations in DNA, RNA, and protein sequences. The HGCN rule includes specific conventions for representing different types of variations, such as substitutions, insertions, deletions, and rearrangements, in a standardized format. This allows researchers and clinicians to accurately communicate and interpret genetic variations across different studies and databases. The HGCN rule plays a crucial role in facilitating the understanding and sharing of genetic information in the scientific community.

- "c." for a coding DNA sequence (like c.76A>T)
- "g." for a genomic sequence (like g.476A>T)
- "m." for a mitochondrial sequence (like m.8993T>C)
- “r.” for an RNA sequence (like r.76a>u)
- ”p.” for a protein sequence (like p.Lys76Asn)

causes of variation:

There are two types of causes, 1= Genetic causes of variation which including: Meiosis, Sexual reproduction, Mutations 2= Environmental causes of variation which including: Nutrients, Drugs, Temperature, Physical training.

Role of positive and negative apistasis:

Positive epistasis occurs when the effect of one gene enhances the effect of another gene, leading to a more pronounced phenotype. Negative epistasis, on the other hand, occurs when the effect of one gene suppresses or diminishes the effect of another gene, resulting in a reduced phenotype.

Point mutation, Transition mutation and transversion mutation:

1= Point mutations are small genetic changes where a single nucleotide is altered in DNA. 2= Transition mutations replace a purine with another purine or a pyrimidine with another pyrimidine, maintaining structure. 3= Transversion mutations switch a purine for a pyrimidine or vice versa, causing a more pronounced structural change. Both transitions and transversions can lead to genetic variation, influencing traits and disease susceptibility.

why we study polymorphism in organism?

Studying polymorphism in organisms is crucial because it reveals the genetic variations that drive diversity within a species. This diversity underlies the ability of organisms to adapt to changing environments, helps identify genetic factors linked to diseases, aids in conservation efforts, and informs agricultural practices for improved crop yields and livestock health.

Mitotic apparatus :

The mitotic apparatus is a cellular structure involved in cell division. It includes spindle fibers and centrosomes that segregate chromosomes, ensuring accurate distribution of genetic material during mitosis.

Six multifactorial diseases:

● Asthma ● Cancers ●Ciliopathies ● Cleft palate ● Diabetes ●Heart diseases.

Diff between Northern , Southern & Western blot:

Southern blotting: Southern blot is a method used in molecular biology for detection of a specific DNA sequence in DNA samples.

Northern blotting: The northern blot is a technique used in molecular biology research to study gene expression by detection of RNA (or isolated mRNA) in a sample.

Western blotting: The western blot (sometimes called the protein immunoblotting) is a widely used analytical technique used to detect specific proteins in a sample of tissue homogenate or extract.

Causes of polyploidy:

Polyploidy is a genetic condition where an organism has more than two sets of chromosomes in its cells. It can occur naturally or be induced artificially.

Natural causes include errors in cell division, resulting in the duplication of chromosomes, while artificial causes involve interventions like chemicals or irradiation to induce extra chromosome sets. Polyploidy can lead to increased genetic diversity, altered physical traits, and sometimes even enhanced adaptability to different environments.

comparison b/w mitosis/ meiosis.

Mitosis:

1. One division
2. Produces two identical diploid cells
3. Maintains the same chromosome number
4. Limited genetic variation
5. Occurs in somatic cells
6. Purpose is cell replication, growth, and tissue repair

Meiosis:

1. Two divisions (meiosis I and II)
2. Produces four genetically diverse haploid cells
3. Reduces chromosome number by half
4. Significant genetic variation through crossing over and independent assortment
5. Occurs in reproductive cells
6. Purpose is the production of gametes for sexual reproduction.

Complications of inheritance pattern:

● Multiple genes - epistasis, polygenic traits ● Genes & the environment - sexinfluenced traits, incomplete penetrance.● A common recessive inheritnace can give dominant pattern ● Autosomal dominant inheritance with nonpenetrance ● Autosomal dominant inheritance with variable expression.

Recessive inheritance diseases:

Recessive inheritance diseases manifest when an individual inherits two copies of a mutated gene. Examples include cystic fibrosis, sickle cell anemia, Tay-Sachs disease, and phenylketonuria (PKU). These conditions typically appear when both parents are carriers of the mutated gene, passing it on to their offspring.

Steps of organic method of DNA extraction:

1. **Cell Disruption:** Cells are broken open to release DNA.
2. **Protein Removal:** Enzymes and detergents eliminate proteins.
3. **DNA Precipitation:** Ethanol or isopropanol is added to cause DNA to precipitate.
4. **Washing:** DNA is washed and purified.
5. **Elution:** DNA is dissolved in a buffer for use in analysis or experimentation.

In which form blood group exist:

Blood groups exist in various forms due to different combinations of antigens and antibodies. The ABO blood group system involves four main blood types: A, B, AB, and O. These types are determined by the presence or absence of antigens A and B on the surface of red blood cells, along with antibodies against the missing antigens. The Rh factor, another important antigen, adds the positive (+) or negative (-) designation to the blood type (e.g., A+, B-, O-). This results in a variety of blood group combinations.

Organic and inorganic method of DNA extraction:

Organic Method: The organic method of DNA extraction involves cell disruption using physical methods like grinding or enzymatic digestion. Proteins are removed with detergents and enzymes, followed by DNA precipitation using organic solvents like ethanol. After washing, purified DNA is eluted in a buffer for further analysis.

Inorganic Method: The inorganic method employs silica columns or magnetic beads coated with silica-binding materials. DNA binds to these surfaces, while contaminants are washed away. Elution releases pure DNA, suitable for downstream applications. This method minimizes organic solvent use and is amenable to automation.

Multiple traits with example:

Multiple traits refer to the simultaneous inheritance of multiple characteristics controlled by different genes. An example is skin color, influenced by multiple genes affecting pigmentation, such as in humans. These traits exhibit complex inheritance patterns due to interactions between genes and environmental factors, contributing to diverse outcomes.

RBC lysis in RNA extraction:

Red Blood Cell (RBC) lysis is a crucial step in RNA extraction from blood samples. It involves breaking down RBCs to release hemoglobin, preventing interference with RNA analysis. Lysis buffers containing detergents and salts are used to rupture RBC membranes. This step ensures the isolation of high-quality RNA from white blood cells, which are then processed for downstream applications like gene expression analysis.

Mitochondrial disease:

Mitochondrial diseases are a group of genetic disorders that affect the mitochondria. Examples: Myoclonic epilepsy • Mitochondrial recessive ataxia • Leber hereditary optic neuropathy • Sensory ataxia neuropathy.

Mandelian and non mandelian genetics:

Mendelian genetics refers to the principles of inheritance proposed by Gregor Mendel, involving the transmission of traits from parents to offspring through discrete units (genes) following predictable patterns. **Non-Mendelian genetics** refers to inheritance patterns that do not strictly adhere to Mendel's laws, including incomplete dominance, co-dominance, polygenic inheritance, and epistasis.

Genetic disease and multifactorial disease:

Genetic diseases result from mutations in a single gene, like cystic fibrosis. **Multifactorial diseases**, such as heart disease, arise from a combination of genetic susceptibility and environmental factors. Their inheritance involves complex interactions between multiple genes and environmental influences.

Role of mitosis and meiosis in animal:

Mitosis ensures cell growth, repair, and asexual reproduction by dividing a cell into two genetically identical daughter cells. **Meiosis** produces haploid gametes for sexual reproduction, enabling genetic diversity through the fusion of gametes with different genetic combinations.

How can we identify chromosomes:

We can identify chromosomes through various techniques, such as karyotyping, fluorescent in situ hybridization (FISH), or chromosomal microarray analysis (CMA). These methods allow us to visualize and analyze the structure and number of chromosomes in a cell.

Eye color, hair, facial features, appendages write recessive and dominant:

1= Eye color:

Dominant: brown eyes

Recessive: grey, green, hazel, blue eyes

2= Hair:

Dominant: non-red hair

Recessive: red hair

3= Facial features:

Dominant: dimples

Recessive: non-dimples

4= Appendages:

Dominant: extra digits

Recessive: normal number

Plants examples triploid tetraploid hexaploid octaploid and decaploid:

Here are examples of ploidy levels in plants:

1. **Triploid:** A plant with three sets of chromosomes. Example: Seedless watermelons are often triploid. They are created by crossing a diploid watermelon with a tetraploid watermelon.
2. **Tetraploid:** A plant with four sets of chromosomes. Example: Durum wheat (Triticum turgidum) is often cultivated in a tetraploid form.
3. **Hexaploid:** A plant with six sets of chromosomes. Example: Bread wheat (Triticum aestivum) is a hexaploid. It's one of the most widely cultivated crops for its edible grains.
4. **Octaploid:** A plant with eight sets of chromosomes. Example: The cultivated strawberry, Fragaria x ananassa, is often octaploid.
5. **Decaploid:** A plant with ten sets of chromosomes. Example: Some varieties of sugar cane can be decaploid.

What is the difference between spermatogenesis and oogenesis?

Spermatogenesis:

1. Occurs in the testes.
2. Results in the production of four functional sperm cells.
3. Begins at puberty and continues throughout the male's life.

4. Produces haploid sperm cells.
5. Meiosis divisions are continuous, resulting in minimal cytoplasmic differentiation.
6. Produces millions of sperm cells daily.

Oogenesis:

1. Occurs in the ovaries.
2. Results in the production of one functional egg cell (ovum) and three polar bodies.
3. Begins before birth and ceases during menopause.
4. Produces haploid egg cells.
5. Meiosis divisions are discontinuous, with extensive cytoplasmic differentiation.
6. Produces one egg cell per month (in fertile women).

Short range applications of molecular markers:

1. **Parentage Determination:** Molecular markers reveal biological parentage in animals or humans.
2. **Genetic Distance Estimation:** Markers quantify genetic relatedness between individuals or populations.
3. **Determination of Twin Zygosity:** Identifies whether twins are fraternal or identical based on genetic markers.
4. **Sexing of Pre-implanted Embryos:** Molecular markers determine the sex of embryos before implantation.
5. **Identification of Disease Carriers:** Markers detect carriers of genetic diseases, aiding in genetic counseling and disease prevention.

Percentage of mono,di, poly allelic in human

1. 30% genes in humans are di-allelic they exist in two forms.
2. 70% are mono-allelic, they exist only in one form.
3. Few are poly-allelic.

Polyploidy and it’s types

Polyploidy is a genetic condition where an organism has more than two sets of chromosomes. Types include autopolyploidy, where multiples of the same genome are present, and allopolyploidy, involving genomes from different species. Polyploidy can enhance genetic diversity and affect an organism's traits and adaptability.

Genetic/inherited variation and it’s two examples:

Genetic or inherited variations refer to the differences in genetic information that are passed down from parents to their offspring. These variations can occur due to changes in DNA sequences, such as mutations or genetic recombination during the formation of reproductive cells.

Some examples of genetic variations include:

1. Eye color 2. Blood type 3. Height: 4. Skin color 5. Inherited diseases.

Autosome dominant , autosome recessive, multifactorial, x linked recessive:

1. **Autosomal Dominant:** A genetic inheritance pattern where a single copy of a dominant allele on an autosome (non-sex chromosome) results in the expression of a trait, even if the other allele is recessive.
2. **Autosomal Recessive:** An inheritance pattern where two copies of a recessive allele on autosomes are necessary for the expression of a trait. If only one copy is present, the trait remains hidden.
3. **Multifactorial:** Traits influenced by both genetic factors and environmental influences. These traits don't follow simple inheritance patterns and result from the interaction of multiple genes and environmental factors.

4. **X-Linked Recessive:** A genetic trait carried on the X chromosome. Males (XY) have a higher chance of expressing these traits, as they have only one X chromosome, while females (XX) need two copies of the recessive allele to show the trait.

Comparison between continuous and non continuous variations:

Continuous variations involve a gradual, uninterrupted range of phenotypic traits within a population, influenced by multiple genes and environmental factors (e.g., height). Non-continuous variations exhibit distinct, discrete phenotypes with limited intermediate forms, often controlled by single genes (e.g., blood type).

Quantitation of RNA.

Quantification of RNA involves measuring the amount of RNA molecules in a sample. Common methods include spectrophotometry, qRT-PCR, and RNA sequencing. The chosen method depends on the purpose and sensitivity required for the analysis.

What are sex chromosomes, What are homogametic and heterogametic

Sex chromosomes are a pair of chromosomes that determine an individual's sex. In humans, they are typically denoted as X and Y. Males have one X and one Y chromosome (XY), while females have two X chromosomes (XX).

Homogametic refers to the sex that has two identical sex chromosomes, either XX (in females) or YY (in certain male animals). **Heterogametic** refers to the sex that has two different sex chromosomes, typically XY (in males). In humans, females are homogametic (XX) and males are heterogametic (XY).

Spontaneous Mutation and mutagenic agent:

Spontaneous mutations occur naturally due to errors in DNA replication or environmental factors. **Mutagenic agents** are external substances that increase mutation rates, like chemicals or radiation. Both contribute to genetic diversity, potentially leading to evolutionary changes and occasionally causing harmful effects in organisms.

write 3 types of genetic disorder:

1. **Cystic Fibrosis:** A hereditary disorder affecting the respiratory and digestive systems, causing mucus buildup.
2. **Down Syndrome:** A chromosomal disorder resulting in intellectual and developmental delays.
3. **Hemophilia:** A genetic condition impairing blood clotting, leading to excessive bleeding and bruising.

Calculate the number of following for specific trait gene if 10 alleles for specific gene Genotype, homogyzous, heterozygous.

For a specific gene with 10 alleles, the number of genotypes can be calculated using the formula n*(n+1)/2, where n is the number of alleles. In this case, with 10 alleles: 10*(10+1)/2 = 55 genotypes. Homozygous genotypes (same alleles) would be 10, as each allele can form a homozygous pair. Heterozygous genotypes (different alleles) would be 45 (55 total genotypes - 10 homozygous genotypes), accounting for various combinations of different alleles.

So, Genotypes=55, Homozygous genotypes= 10, and Heterozygous genotypes= 45

Prokaryotic and eukaryotic chromosomes:

Prokaryotic chromosomes lack a nuclear membrane, existing as a single circular DNA strand in the cytoplasm, often accompanied by plasmids. Eukaryotic chromosomes are linear DNA molecules enclosed within a nucleus, along with histone proteins, and can vary in number. Eukaryotes also exhibit complex DNA packaging and undergo mitosis and meiosis, whereas prokaryotes divide through binary fission.

Down synonym:

Down syndrome, also known as Trisomy 21, is a genetic disorder caused by the presence of an extra copy of chromosome 21. This condition leads to cognitive and developmental delays, as well as distinctive physical features. Individuals with Down syndrome often have a range of abilities and challenges, requiring varying levels of support. Early intervention, specialized education, and medical care can help them lead fulfilling lives. Emphasizing inclusion and understanding is vital for fostering a more inclusive society.

Is there any difference between telophase 1 and teleophase 2 of meiosis? please discuss:

Yes, there is a difference between telophase I and telophase II of meiosis. Telophase I follows the first division of meiosis and involves the separation of homologous chromosomes, while telophase II follows the second division and involves the separation of sister chromatids.

Clinical signs of down syndrome:

some clinical signs often associated with Down syndrome: 1. Distinct facial features 2. Hypotonia (low muscle tone) 3. Developmental delays 4. Cognitive impairment 5. Heart defects 6. Gastrointestinal issues 7. Hearing and vision problems 8. Thyroid issues 9. Short stature 10. Speech and language difficulties.

Loss of function and gain of function:

Loss of function refers to a mutation causing a gene to lose its normal function, potentially leading to reduced or absent protein activity. Gain of function results from mutations enhancing gene activity, often causing the protein to gain new or increased functions, potentially leading to abnormal cellular processes or diseases.

Explain the experiment performed by T.H Morgan with and example:

T.H. Morgan conducted an experiment using fruit flies (Drosophila melanogaster). He crossed a female fly with red eyes (XX) with a male fly with white eyes (XY), producing all offspring with red eyes. When he crossed these offspring, he unexpectedly found that only male flies had white eyes, while females all had red eyes.

This observation led Morgan to conclude that the gene for eye color was located on the X chromosome and that the Y chromosome didn't carry the same gene. This groundbreaking discovery helped establish the concept of sex-linked inheritance and provided key insights into the role of chromosomes in passing on genetic traits.

EXCEPTIONS TO MENDELIAN INHERITANCE:

1= Incomplete dominance 2= Codominance 3= Multiple alleles 4= Polygenic traits 5=Epistasis 6= Pleiotropy 7= Environmental effects on gene expression 8= Linkage 9= sex linkage.

Difference between prokaryotic and eukaryotic chromosome:

Prokaryotic chromosomes, located in the nucleoid region, are circular and less compact than eukaryotic counterparts due to the lack of histones. Each prokaryotic cell holds one chromosome with simple organization and minimal non-coding DNA. Replication starts at a single origin, followed by binary fission for cell division.

Eukaryotic chromosomes, within the nucleus, are linear and heavily compacted by histones. Eukaryotic cells possess multiple chromosomes with intricate organization, encompassing substantial non-coding DNA. Replication involves numerous origins and division through mitosis or meiosis. Eukaryotic chromosomes are larger, contain introns, and often feature single-cistronic genes with fewer operons.

Define nucleotide, nucleoside, purine, and pyrimidine.

A nucleotide is a building block of nucleic acids (DNA and RNA) consisting of three components: a sugar (ribose or deoxyribose), a phosphate group, and a nitrogenous base. A **nucleoside** is a similar structure, but lacks the phosphate group. **Purines** (adenine and guanine) and **pyrimidines** (cytosine, thymine, and uracil) are the two types of nitrogenous bases. Purines

have a double-ring structure, while pyrimidines have a single-ring structure. Nucleotides combine to form the backbone of DNA and RNA strands.

Forward, reverse and neutral mutation

1. Forward Mutation: Alters one DNA base, possibly changing a protein's amino acid sequence.

2. Reverse Mutation: Mutated gene reverts back to its original sequence, restoring function.

3. Neutral Mutation: Doesn't impact an organism's traits or function noticeably.

Write about variation in autosomal dominant inheritance in reference of following terms. Penetrance, expressivity, pleiotropy.

Penetrance: Penetrance indicates the likelihood of a dominant gene carrier showing the related trait. High penetrance means most carriers display it; low means fewer do, like in incomplete penetrance genetic disorders.

Expressivity: Expressivity refers to the trait's intensity or variation among individuals with the same dominant gene. For example, autosomal dominant polydactyly shows differing extra finger or toe numbers due to variable expressivity.

Pleiotropy: Pleiotropy involves a single gene mutation affecting multiple unrelated traits. In autosomal dominant disorders with pleiotropy, one gene causes various effects. Marfan syndrome's FBN1 gene mutation leads to tallness, heart defects, and joint hypermobility.

Precautions during extraction of RNA:

Use RNase-free equipment and reagents. • Wear gloves to prevent skin contamination. • Work in a clean area to avoid contamination. • Consider RNA stabilizing agents for storage. • Minimize mechanical shearing of RNA. • Avoid exposing RNA to high temperatures. • Homogenize carefully for even RNA distribution. • Process samples quickly to prevent degradation. • Prevent cross-contamination between samples. • Follow established RNA extraction protocols. • Perform quality checks after extraction. • Keep detailed documentation of the proc.

Write down the method that converts euchromatin to heterochromatin:

One method that converts euchromatin to heterochromatin is through histone modifications, specifically histone methylation and acetylation. These modifications can alter the chromatin structure, making it more condensed and less accessible for gene expression.

Define heterozygous dominant trait with example.

Heterozygous dominant trait occurs when an individual carries two different alleles for a trait, where the dominant allele's effect is expressed while the recessive allele remains hidden. **For instance**, a person with "Ww" genotype shows a widows peak due to the dominant "W" allele.

Mandelian and non mandelian genetics:

Mendelian genetics refers to the principles of inheritance proposed by Gregor Mendel, involving the transmission of traits from parents to offspring through discrete units (genes) following predictable patterns. **Non-Mendelian genetics** refers to inheritance patterns that do not strictly adhere to Mendel's laws, including incomplete dominance, co-dominance, polygenic inheritance, and epistasis.

Intergenic and Intragenic Mutations:

Intergenic mutations occur in the non-coding regions of DNA between genes, while **intragenic mutations** occur within a gene's coding or non-coding regions. Intergenic mutations often have less direct impact on gene function, while intragenic mutations can affect a gene's structure or function, potentially leading to genetic disorders.

Drawbacks of mendel law of independent assortment:

Drawbacks of Mendel's Law of Independent Assortment: 1= Genetic linkage violates independence. -2= Exceptions due to incomplete dominance, etc. 3= Complex traits involve multiple alleles and genes. 4= Crossing over generates new allele combinations. 5= Doesn't cover inheritance of sex-linked traits. 6= Ignores gene interactions like epistasis. 7= Limited to model organism's traits. 8= Ignores environmental influences. 9= Incomplete penetrance and variable expressivity not considered.

how mutation alter the gene product?

Mutations change a gene's DNA sequence, altering the instructions for making a protein. This can lead to incorrect protein structure, function, or levels, affecting normal cellular processes and potentially causing genetic disorders or diseases.

Explain the haploid- diploid sex determination system:

In the haploid-diploid sex determination system, males develop from unfertilized eggs (haploid, containing half the usual genetic material) and are thus haploid. Females develop from fertilized eggs (diploid, with the full genetic material), making them diploid. This system is observed in certain insects, like bees and ants. It results in a sex ratio of 1:1 as females inherit half their genetic material from their fathers (haploid) and the other half from their mothers (diploid).

Long term impacts of genetics:

1= Healthcare: Personalized medicine, disease prediction, and drug development. 2= Evolution: Natural selection, speciation, and genetic diversity. 3= Agriculture: Genetically modified crops, improved breeding programs. 4= Forensics: DNA identification, tracing human migration. 5= Ethics: Genetic privacy, ethical concerns in genetic engineering. 6= Scientific Advancement: Fundamental research, technological innovation. 7= Conservation: Biodiversity protection, endangered species conservation. 8= Understanding Life: Molecular biology, uncovering life's mechanisms.